

Synopsis: A Milky Way Realization of the Hysterical Response Halo

What the theorems and experiments say (without the proofs)

Andrew Bruce Boyd

September 2026

Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the Milky Way application of the Hysterical Response Halo. It assumes the response-theory and galactic-modelling papers. Each section says *why* the claim matters, then states the result. Proofs and experimental detail live in the full paper.

1 Why a Galactic realization

The galactic-modelling paper supplies $g_{\text{resp}} = CQ/r$, a BTFR attractor, and an asymptotic oblate quadrupole. Those are scaling theorems, not a fitted Galaxy. The present paper asks what happens when the same laws are inserted into a fixed McMillan-type Milky Way baryonic model and compared with published Gaia-based constraints—without fitting a dark-halo density profile.

2 Axisymmetric radial response

Paper 2's $g_{\text{resp}} = CQ/r$ integrated against a razor-thin axisymmetric production surface Σ_J yields a mid-plane double integral. A classical thin-disc angular identity (literature: contour evaluation) collapses it to enclosed production only.

Proposition 2.1 (Axisymmetric radial response).

$$g_R(R) = \frac{CQ(< R)}{R}, \quad Q(< R) = 2\pi \int_0^R R' \Sigma_J(R') dR'.$$

For an exponential source this is $V_{\text{resp}}^2(R) = V_Q^2 \mathcal{T}_d(R/R_d)$ with $\mathcal{T}_d(x) = 1 - (1+x)e^{-x}$ (cumulative kernel: literature / classical incomplete-Gamma identity). A mass-weighted multi-component McMillan-type source gives $\mathcal{T}(R_0) = 0.822$ at $R_0 = 8.2$ kpc.

3 Solar-circle prediction

Paper 2 fixes the BTFR exponent and the zero point $a_0 = H_0 c / 2\pi$. Using the numerical BTFR calibration $a_0 = 1.20 \times 10^{-10} \text{ m s}^{-2}$ (equivalently that formula at $H_0 \simeq 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and $M_b = 6.65 \times 10^{10} M_\odot$,

$$V_Q = 180.4 \text{ km s}^{-1}.$$

With the adopted baryonic speed $V_b(R_0) = 176.5 \text{ km s}^{-1}$,

$$V_c(R_0) = \sqrt{V_b^2 + V_{\text{resp}}^2} = 240.6 \text{ km s}^{-1},$$

compared with the published Cepheid value $236.8 \pm 0.8 \text{ km s}^{-1}$.

4 Radial curve vs published Cepheid bins

Against Feng et al.'s 12 published bins ($6 < R < 18 \text{ kpc}$), the BTFR-normalized smooth realization has RMS residual 6.25 km s^{-1} . It does *not* reproduce the observed dip-and-bump structure, and is not statistically adequate under the quoted bin errors. The empirical MOND/RAR interpolation with the same baryons has RMS 20.1 km s^{-1} .

5 Oblate geometry

The galactic-modelling quadrupole specializes to an exponential disc as follows.

Corollary 5.1 (Exponential-disc Solar estimate).

$$q_{\text{eff}}(R) \simeq \left[1 + 6(R_d/R)^2\right]^{-1/2}.$$

With $R_d = 2.5 \text{ kpc}$ and $R_0 = 8.2 \text{ kpc}$ one has $q_{\text{th}} \simeq 0.80$.

6 Vertical density cross-check

From Paper 2's Poisson closure with an oblate logarithmic response potential, near-plane $K_z \simeq V_{\text{resp}}^2 z / (q^2 R_0^2)$ identifies an equivalent density

$$\rho_{\text{eff}}(q) = \frac{V_{\text{resp}}^2(R_0)}{4\pi G q^2 R_0^2}.$$

Holding the radial response fixed at the Solar-normalized value and using the published local density $0.0117 \pm 0.0035 M_\odot \text{ pc}^{-3}$ gives a geometric fit $q_{\text{fit}} = 0.766$ and 1σ -equivalent interval $0.672 < q < 0.915$. At $q_{\text{th}} = 0.80$ the predicted density is only 0.29σ from the reported central value.

7 What is and is not claimed

Baryonic parameters and $V_b(R_0)$ are inputs. $V_Q^4 = G a_0 M_b$ with $a_0 = H_0 c / 2\pi$ is the galactic-modelling zero-point theorem (numerical a_0 from BTFR/ H_0 calibration). The axisymmetric radial reduction after the literature angular kernel, and the exponential $\mathcal{T}_d$ algebra after the literature cumulative identity, are Lean-checked. Radial χ^2 and q_{fit} are diagnostics, not proofs. A raw Gaia catalogue re-reduction is deferred; the full paper uses published summary constraints.